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Workforce grouping and assignment with learning-by-doing and knowledge transfer

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We consider a workforce allocation problem in which workers learn both by performing a job and by observing the performance of and interacting with co-located colleagues. As a result, an organisation can benefit from both effectively assigning individuals to jobs and grouping workers into teams. A challenge often faced when solving workforce allocation models that recognise learning is that learning curves are non-linear. To overcome this challenge, we identify properties of an optimal solution to a non-linear programme for grouping workers into teams and assigning the resulting teams to sets of jobs. With these properties identified, we reformulate the non-linear programme to a mixed integer programme that can be solved in much less time. We analyse (near-)optimal solutions to this model to derive managerial insights.

Keywords: knowledge transfer; learning curves; integer programming; worker assignment; productivity management

1. Introduction

Organisations can build and maintain competitive advantage through knowledge generation and transfer (Osterloh and Frey 2000). One distinction researchers have drawn is between knowledge that is *explicit*, meaning it can be coded and disseminated without the involvement of the knowledge holder, and that which is *tacit*, meaning it cannot (Johnson-Laird 1983; Nonaka 1994; Polanyi 2009). Instead, as Polanyi (2009) states regarding tacit knowledge, 'we can know more than we can tell.' However, tacit knowledge need not be trapped within the knowledge holder. Instead, it can be shared through the social interaction that occurs when individuals are organised into teams (Nonaka 1994).

In many work settings, individuals are organised into teams, wherein they perform related jobs in such a manner that they can share or transfer knowledge. In manufacturing, Renault credits the team's success in designing and building thousands of new components for the Formula One Red Bull Racing team each year to knowledge transfer among teammembers (Economist Intelligence Unit 2007). In health care, a new technician in a medical laboratory is often paired with an experienced technician for training and learning purposes (Personal Communication with Wendy Voigt, Phlebotomy Services, Department of Pathology, University of Iowa Healthcare, 1 September 2015). Finally, in new product/process development, knowledge transfer has been shown to improve the performance of virtual teams (Soule and Applegate 2005). Regardless of setting, managers can amplify the benefits of knowledge transfer with the teams they organise.

In this paper, we focus on a manufacturing setting. Specifically, we consider parallel production systems wherein equipment is located according to function, as in a job shop. As an example, consider a custom furniture factory wherein lathes are located together. We call such co-located equipment a job group. The manufacturer seeks to organise their employees into teams, which we call worker groups, and then assign each group of workers to a different job group. We study a situation wherein the manufacturer explicitly recognises that an individual's productivity on a job can improve due to learning. In addition, there are two factors under managerial control that impact employee learning: (1) an employee's own experiences performing jobs, and, (2) the transfer of knowledge derived from the experiences of other employees performing jobs in that same group. Ultimately, the organisation seeks to make these worker grouping and assignment decisions to maximise the throughput of the manufacturing system.

Nembhard and Bentfouet (2015) were the first to consider knowledge transfer in the context of worker grouping and assignment, and present an adaptation of the hyperbolic learning curve proposed by Mazur and Hastie (1978), wherein a worker's experience is a function of both personal experience and transferred knowledge. However, as the learning curve is non-linear, the resulting mathematical programming formulation of the worker grouping and assignment problem with knowledge transfer is a Mixed Integer Non-linear Programming (MINLP) problem, rendering it computationally challenging

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to solve. As a result, [Nembhard and Bentefouet \(2015\)](#) presents heuristics for solving this problem. They also consider the selection of workers to employ, a decision we do not optimise.

Instead, in this paper, we analyse the non-linear worker grouping and assignment optimisation problem. This problem encompasses three decisions. First, we need to assign workers to a group. Then, we need to assign this group to a job group. Finally, we need to assign individuals in the group to individual jobs in the jobs group. We prove that there exists an optimal solution in which a worker is assigned to the same group, the group to the same job group, and each worker in the job group to the same job throughout the horizon. This observation enables multiple contributions to production research. From a managerial perspective, it implies that workforce grouping and assignment does not require regular management involvement. Instead, it need only be optimised when the system changes (e.g. the workforce or equipment set changes). In a sense, workforce grouping and assignment can be thought of as a tactical planning decision, instead of an operational one.

From a managerial and algorithmic perspective, we present a method for optimising worker grouping and assignment. Specifically, the observation that a group of workers can specialise on a group of jobs enables the reformulation of the above-mentioned MINLP to a Mixed Integer Programming (MIP) formulation, of which far larger instances can be solved with off-the-shelf optimisation software. We validate the use of this reformulation with a computational study wherein we compare solutions to the worker grouping and assignment problem that result from solving our proposed reformulation with those produced by the heuristics presented in [Nembhard and Bentefouet \(2015\)](#). We see that the solutions produced by solving the reformulation are of much higher quality than those produced by the heuristics. Finally, from a managerial perspective, we analyse this set of high-quality solutions to derive new insights into how grouping should be done in cases wherein solving the reformulation is not an option. Specifically, we show that it is best to group together workers who have dissimilar learning characteristics.

The remainder of this paper is organised as follows. Section 2 is devoted to literature review of the impact of individual learning and knowledge transfer on system performance. Section 3 presents a MINLP formulation of the problem. Section 4 then reformulates this MINLP to a MILP problem. Section 5 presents the experimental design and summarises the results of the computational study, including insights gained from analysing solutions. Lastly, Section 6 concludes the paper.

2. Literature review

In this section, we summarise the relevant literature regarding individual human learning, as well as team formation and knowledge transfer.

2.1 Individual human learning

The impact of experience on production times is often called ‘learning’ in the literature. There exists an extensive body of literature that develops mathematical representations for quantifying the improvement in service and production times as experience increases. These representations are often called learning curves. With the introduction of his power model, [Wright \(1936\)](#) is often credited with introducing the first mathematical description of the relationship between experience and productivity. Since that time, many authors have made additions to the literature. [Dar-El \(2000\)](#), [Jaber and Sikström \(2004\)](#), [Jaber \(2006\)](#), and [Anzanello and Fogliatto \(2011\)](#) provide broad and thorough surveys of the subject. [Dar-El \(2000\)](#) provides a comprehensive review of both learning and forgetting models as well as parameter estimation for learning models developed before 2000.

Much of the existing work in workforce assignment that models individual learning curves is limited due to the non-linear nature of the curves. Recently, for many learning curves for which work experience can be measured in discrete units, [Hewitt et al. \(2015\)](#) present an exact linear reformulation of the learning curves. Results demonstrate the ability to solve problems that are significantly larger than those found in the literature. [Jin, Thomas, and Hewitt \(2016\)](#) applies the reformulation, introduces additional model strengthening techniques, and develops a method for finding an initial feasible solution for the problem of minimising the makespan in a job shop environment when employs learn. Like [Hewitt et al. \(2015\)](#), [Jin, Thomas, and Hewitt \(2016\)](#) solves problems to optimality and solves much larger problems than those found in the literature. This paper linearises the integer programmes by taking advantage of structure specific to this problem rather than adapting the techniques introduced in [Hewitt et al. \(2015\)](#). Additional workforce planning literature that explicitly models individual human learning includes [Corominas, Olivella, and Pastor \(2010\)](#), [Olivella, Corominas, and Pastor \(2013\)](#), and [Heimerl and Kolisch \(2010\)](#). A comprehensive review of research that incorporates learning in workforce assignment models can be found in [Hewitt et al. \(2015\)](#).

2.2 Team formation and knowledge transfer

Team formation and performance in manufacturing settings has been studied (Banker et al. 1996; Banker, Field, and Sinha 2001; Fitzpatrick and Askin 2005), often in the context of cellular manufacturing (Askin and Huang 2001; Norman et al. 2002; Sengupta and Jacobs 2004). Most relevant to this paper, there have been investigations into how and why working in a team environment can improve individual performance. Faraj and Sproull (2000) and Lewis (2003) suggest that teams composed of individuals who have experience working together are more accurate and have a collective sense of each individual's expertise. Similarly, Uzzi and Lancaster (2003) discuss how trust can promote the exchange of 'private' knowledge and information.

However, there is little research that reflects the transfer of knowledge to a worker from his/her colleagues when forming teams. This is in part because until recently, a mathematical description of knowledge transfer did not exist. Nembhard and Bentefouet (2015) propose such a description, which is an extension of the three-parameter hyperbolic learning curve proposed in Mazur and Hastie (1978). In the learning curve proposed by Nembhard and Bentefouet (2015), a worker's cumulative experience is a function of both direct and indirect experience. Nembhard and Bentefouet (2015) embed this learning curve into an optimisation model that seeks to jointly make three decisions: (1) selecting a workforce, (2) grouping the workforce into teams and (3) assigning the teams of workers to groups of jobs, with the goal of maximising system throughput. However, due to it being challenging to solve computationally, the authors decompose this problem into three separate optimisation problems, one for each decision. They then present heuristics to solve each of these three problems.

Our work builds on the work of Nembhard and Bentefouet (2015), with preliminary findings and results found in Jin (2016). Notably, we seek to provide exact solutions for a specific case in their work. Specifically, we demonstrate how to solve exactly, without decomposition, the grouping and job assignment portion of the parallel production problem. We note that we provide a more general model than what is allowed by the decomposition proposed in Nembhard and Bentefouet (2015). The previously proposed decomposition fixes a worker's group over the horizon. Our formulation allows each worker to switch groups in each period. However, we also show that there exists an optimal solution such that workers will not switch groups. This result allows us to solve to optimality larger problems than the previous work. We also compare our exact solutions to solutions generated by best heuristics presented Nembhard and Bentefouet (2015) for the grouping and assignment of workers in parallel production systems. We note that we use a different set of test instances from those used in the literature, though we generate the instances in a similar fashion. In contrast to existing results, our results show that dissimilar workers, workers who differ in learning rates, initial experience, and asymptotic productivity, should be grouped. These results demonstrate the value that we can gain when we can exactly solve a problem and in doing so without decomposition.

3. Problem setting and model

In this section, we first present a description of the problem studied and then review the quantitative model of learning that incorporates knowledge transfer that we use in the paper. Finally, we present a mixed integer non-linear programming model of the problem.

In this paper, we consider an organisation that seeks to develop a workforce plan for a parallel system over a finite planning horizon that is divided into discrete periods. For example, the organisation seeks to determine a workforce plan for each day (the discrete period) of the next month (the planning horizon). In this parallel system, jobs have already been partitioned into groups that require the same skills, and jobs of the same group are in proximity in the production facility to facilitate the transfer of knowledge. The goal of the organisation is to assign workers to groups, and then groups of workers to groups of jobs, in order to maximise total productivity throughout the planning horizon. We consider a setting wherein each worker can be assigned to at most one worker group at a time and there is a limit on the size of the worker group assigned to a job group.

3.1 Quantitative model of human learning that includes knowledge transfer

As noted in the previous section, Nembhard and Bentefouet (2015) propose a mathematical description of knowledge transfer. In this section, we provide a detailed description of that learning curve and then discuss how we implement the learning curve in our math programme in the next section. Let D (direct experience) represents the number of times a person has performed a job. Let S (indirect experience) represents the experience of others working on jobs that are in close enough proximity for an individual to learn from their work, either by direct observation or social interaction. However, not all indirect experience is transferred. Instead, S is scaled by a parameter θ (a percentage) that represents an individual's ability to learn via transfer. Specifically, θ represents what fraction of indirect experience is transferred to the individual.

The learning function makes use of three additional parameters. The parameter K is the asymptotic limit of production output. This parameter implements the notion common in the learning literature that there is an upper limit on each individual's

productivity regardless of how experienced that individual might be. Parameter r is a measure of the individual's ability to learn expressed as the level of experience need to achieve an output of $K/2$. The learning rate is a measure of how fast an individual learns from the experience that he/she gains. The parameter p is a worker's prior experience, and is measured in the same units as D . In practice, all of the given parameters would need to be derived from data. Such data could be captured during training, on the job, and from each workers' work history. See [Shafer, Nembhard, and Uzumeri \(2001\)](#) for further discussion.

Given these parameters, the model of learning by direct and indirect experience is as follows:

$$\varphi = K \left(\frac{\theta \times S + D + p}{\theta \times S + D + p + r} \right). \quad (1)$$

As we will show later, there is an optimal solution to our planning problem wherein a worker works on the same job group throughout the planning horizon. As such, the fact that we do not model forgetting is not a limitation of the model.

3.2 Mathematical programming model

In this section, we present the formal math programming formulation for the problem. For the presented model, we restrict ourselves to the case in which every job in each group requires the same set of skills. We make this assumption to align our model with the instances that we use to test the model in Section 5 as well as to facilitate comparisons with the methods in the literature. We provide a more general formulation in Appendix 1.

To model the problem, we assume a horizon of T periods, and let the set $\mathcal{T} = \{1, 2, \dots, T\}$ index the set of periods in the horizon. We define $\mathcal{I} = \{1, 2, \dots, I\}$ to be a set of workers. In each period t , each worker is assigned to a job group in $\mathcal{G} = \{1, 2, \dots, G\}$. We let x_{ig}^t be a binary variable indicating whether or not worker i is assigned to job group g in period t . We assume that each job group g in $\mathcal{G}$ has a limit L_g on the number of workers that can work job g in any given period.

Associated with each pair of a worker and a job are the learning parameters that determine the worker's ability to learn the job. As each job group g consists of jobs that require the same skills, we model an individual's ability to learn at a job group level. As such, we let K_{ig} and p_{ig} denote the asymptote parameter and prior experience of worker i on job group g . The parameter r_{ig} captures the rate at which worker i learns jobs in group g by expressing how much experience the individual would need to achieve an output level of $K_{ig}/2$. We note that with these parameters we are explicitly modelling that the workforce is heterogeneous with respect to productivity and learning. Finally, associated with each worker is a transfer parameter $\theta_i \in [0, 1]$, which represents the fraction of knowledge transferred to worker i from her/his coworkers that are working on jobs in the same job group.

For each period t in the horizon, we let c_{ig}^t denote the accumulated work experience of worker i in job group g up to t . Then, for worker i on job group g in period t , we implement Equation (1) as:

$$\varphi_{ig}(c_{ig}^t) = K_{ig} \frac{c_{ig}^t + p_{ig}}{c_{ig}^t + p_{ig} + r_{ig}}. \quad (2)$$

As discussed previously, the accumulated work experience c_{ig}^t is a function of direct and indirect experience. The direct experience is the number of periods in which worker i worked job group g by time t , i.e. $D_{ig}^t = \sum_{t' < t} x_{ig}^{t'}$. The indirect experience for worker i in period t is the experience of colleagues that impacts i 's productivity on job group g . We denote the indirect experience as S_{ig}^t , and compute

$$S_{ig}^t = \sum_{i' \in \mathcal{I}: i' \neq i} \sum_{t' < t} \varphi_{i'g}(c_{i'g}^{t'}) x_{i'g}^{t'}. \quad (3)$$

Then, we calculate the total accumulated experience of worker i on job group g in period t as:

$$c_{ig}^t = \theta_i S_{ig}^t + D_{ig}^t. \quad (4)$$

The objective is to maximise the total system production during the horizon (which we often refer to as system throughput). We let the continuous variable o_{ig}^t represent worker i 's output on job g in period t . We define the following non-linear programme for assigning workers to job groups in a parallel system:

$$\begin{aligned} & \text{Max} \sum_{i \in \mathcal{I}} \sum_{g \in \mathcal{G}} \sum_{t \in \mathcal{T}} o_{ig}^t \\ & \text{s.t.} \quad (2) - (4) \end{aligned}$$

$$\sum_{i \in \mathcal{I}} x_{ig}^t \leq L_g, \quad g \in \mathcal{G}, t \in \mathcal{T} \quad (5)$$

$$(PM) \quad \sum_{g \in \mathcal{G}} x_{ig}^t \leq 1, \quad i \in \mathcal{I}, t \in \mathcal{T} \quad (6)$$

$$o_{ig}^t \leq x_{ig}^t \phi_{ig}(c_{ig}^t), \quad i \in \mathcal{I}, g \in \mathcal{G}, t \in \mathcal{T} \quad (7)$$

$$x_{ig}^t \in \{0, 1\}, \quad i \in \mathcal{I}, t \in \mathcal{T}, g \in \mathcal{G}. \quad (8)$$

Constraints (5) restrict the number of workers who can work in each job group in each period. Constraints (6) enforce that a worker is assigned to at most one group. Constraints (7) indicate that the output associated with a job group in a period is limited both by each worker's productivity and whether the worker worked that job group in that period. Constraints (8) specify the domain of the x variables. We refer to the model as 'PM'.

4. Reformulation of the MINLP

Given the state of today's off-the-shelf optimisation solvers, the non-linear nature of PM renders it computationally challenging to solve for instances of moderate-to-large size. However, by analysing the structure present in optimal solutions to PM, we can reformulate it as a linear, mixed integer programme. Specifically, we extend the analysis of Nembhard and Bentefouet (2012), who show that, in a parallel system and under some conditions, the optimal schedule of a worker task assignment problem with individual learning is a schedule with specialised workers. In other words, workers perform the same job, in our case work the same job group, during the planning horizon. However, the result in Nembhard and Bentefouet (2012) does not address a situation wherein knowledge transfer occurs. We show that the result also extends to situations wherein there is knowledge transfer.

We formally state the result in Proposition 4.1. The proof of the result can be found in Appendix 2. We present a proof for a more general formulation of the problem than we presented in the previous section. Notably, in the proof, we allow for the case in which the jobs in each group differ, but that knowledge transfer is still possible. We present a model for this more general problem in Appendix 1.

Proposition 4.1 *In parallel systems in which workers have monotonically non-decreasing learning functions on jobs, the optimal schedule is a schedule with specialised workers. In other words, no workers switch jobs or job groups within the horizon and workers are assigned to a single task.*

The conclusion of Proposition 4.1, that neither worker groupings nor group assignments need change during the planning horizon, enables us to simplify PM in the following ways:

- (1) We need not model that decisions may change from one period (t) to the next ($t + 1$). In other words, we need not model with variables that are indexed by time as there is an optimal solution, $\bar{x}$, wherein $\bar{x}_{ig}^t = \bar{x}_{ig}^{t'}$, $\forall i \in \mathcal{I}, g \in \mathcal{G}, t, t' \in \mathcal{T}, t \neq t'$. Instead, we can model the choice of one set of worker groups and group assignments that will remain in place for the entire planning horizon, $\mathcal{T}$.
- (2) Enumerating all possible worker groups and group assignments, and associating a decision variable with each enumerated item, enables the calculation of the production rate of each worker on each job with Equation (2) to be done in a data-processing step as opposed to in the context of solving a MINLP. Because such decisions need not change from one period to the next, we need one variable for each enumeration. If that was not the case, we would need $|\mathcal{T}|$ such variables, which would limit the size of instance that could be solved.

As such, consider a set of workers $w = \{i_1, \dots, i_n\}$, that are all assigned to job group g . We can calculate the throughput of this worker group w working job group g over the entire planning horizon as :

$$O_w^g = \sum_{l=1}^n \sum_{t \in \mathcal{T}} K_{ilg} \frac{c_{ilg}^t + p_{ilg}}{c_{ilg}^t + p_{ilg} + r_{ilg}}. \quad (9)$$

We let the set W_g denote the set of all potential worker groups w that can be assigned to job group g . With such an enumeration, we can reformulate PM to a linear, mixed integer programme. Specifically, we introduce the binary variable z_w^g to indicate whether the group of workers $w \in W_g$ is assigned to the job group g . The resulting reformulation, which can be seen as a set packing problem, is as follows:

Table 1. Mean and variance–covariance matrix for K , p , and r .

$\mu = \begin{bmatrix} 1.448 \\ 1.963 \\ 2.0446 \end{bmatrix}$		$\log(K)$	$\log(K)$	$\log(p)$	$\log(r)$	
		$\log(p)$				$\log(K)$
		$\log(r)$				$\log(p)$
						$\log(p)$
			$\Sigma = \begin{bmatrix} 0.0045 & 0.0167 & 0.0191 \\ 0.0167 & 0.4621 & 0.2443 \\ 0.0191 & 0.2443 & 0.1905 \end{bmatrix}$			

$$\begin{aligned} \text{Max} \quad & \sum_{g \in \mathcal{G}} \sum_{w \in W_g} O_w^g z_w^g \\ \text{(RPM)} \quad & \sum_{w \in W_g} z_w^g \leq 1, \quad g \in \mathcal{G} \end{aligned} \quad (10)$$

$$\sum_{g \in \mathcal{G}} \sum_{w \in W_g: i \in w} z_w^g \leq 1, \quad i \in I \quad (11)$$

$$z_w^g \in \{0, 1\}, \quad w \in W_g, g \in \mathcal{G}. \quad (12)$$

As with the original formulation, the objective maximises the throughput of the system. Constraints (10) ensure that at most one group is assigned to each job group. Similarly, Constraints (11) ensure that each worker is assigned to at most one group of workers. Finally, Constraints (12) enforce binary decision variables.

5. Computational study

In this section, we describe the computational study performed to address the following issues/questions. First, we validate the use of a model that explicitly recognises that knowledge transfer occurs by comparing the system throughputs enabled by plans that were derived with recognition of knowledge transfer with plans that were derived without. Second, we assess the capabilities of solving the reformulated MIP (RPM). To do so, we first benchmark the quality of the solutions it produces against those produced by the heuristic policies presented in [Nembhard and Bentefouet \(2015\)](#). Then, we perform a computational study to determine how large an instance of the RPM we can solve within a given time limit. Third, we study optimal solutions to the RPM to derive managerial insights. We next describe the details of our computational experiments.

5.1 Experimental settings

All results presented in this section were derived from experiments that were performed on a cluster of computers wherein each node has 16 2.6 GHz cores and 4 GB of memory. All instances were solved using Gurobi version 6.5, and all Gurobi parameters were left at their default values.

Each instance in the data-set includes a workforce, wherein a worker is defined by his/her learning traits (asymptote productivity rate K , learning rate r , prior experience p , and knowledge transfer parameter θ). In all instances, the first three traits are determined for each individual in a workforce by drawing from normal univariate distributions with the means and variance–covariances given in Table 1 (first presented in [Jin \(2016\)](#)). We note that these distribution parameters have been used in are originally presented in [Shafer, Nembhard, and Uzumeri \(2001\)](#) and are used to create instances in the existing literature on knowledge transfer. The transfer parameter θ is generated using a normal distribution with mean 0.664 and variance 0.409. We used only positive values generated from this distribution for our instances.

Throughout much of our computational study, we consider instances wherein the number of jobs and workers is the same, and the bounds on the number of workers that can be assigned to each job group are the same, i.e. $L_g = L_{g'}$.

5.2 The impact of knowledge transfer on throughput

We first study the extent by which recognising that knowledge transfer occurs when determining workforce groups and plans impacts throughput. We study this on a set of 100 instances, each with 9 workers and 9 jobs. The nine jobs are in parallel and are grouped into three job groups. For each instance, we generate two solutions (groups of workers and assignments of worker groups to job groups) and compare the throughput yielded by each solution.

The first solution is derived by solving RPM. As such, this solution is derived when explicitly recognising that knowledge transfer occurs. We refer to the throughput associated with such a solution as obj_{kt} . The second solution is derived in a manner that mimics when knowledge transfer is not considered when deriving a workforce plan. To derive such a solution,

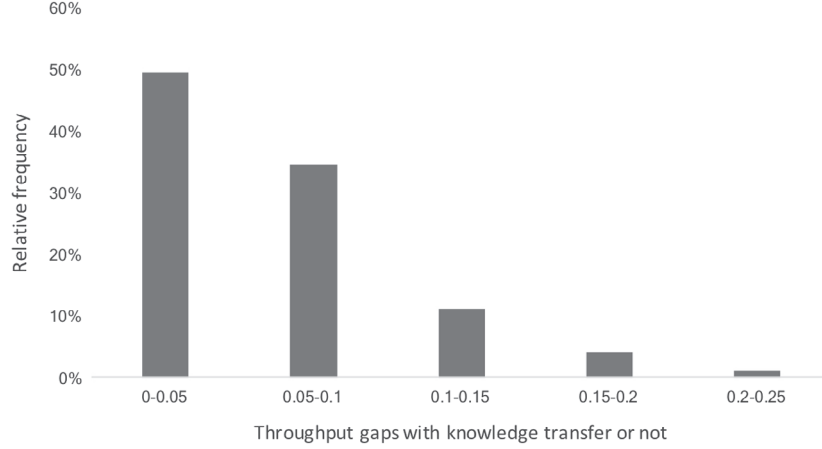


Figure 1. Distribution of throughput gaps when recognising knowledge transfer and not.

we again solve RPM but do so after setting each individual's knowledge transfer parameter (θ_i) to zero. Setting $\theta_i = 0, i \in \mathcal{I}$ essentially turns off recognition of knowledge transfer when deriving the solution. However, knowledge transfer does occur and can impact throughput even if it is not recognised when deriving the plan. Thus, to evaluate the throughput, we evaluate this solution in RPM, albeit with the knowledge parameter for each individual set to the original parameter value. We refer to this throughput as obj_{no-kt} . We calculate the gap in the two throughputs for each instance as $(obj_{kt} - obj_{no-kt})/obj_{kt}$.

We illustrate the distribution of these gaps in Figure 1 and note that over the 100 instances solved, the average gap in throughput is 5.8%. Both the distribution and the average indicate that recognising that knowledge transfer occurs when determining a grouping of workers and assignment of groups of workers to job groups can have a significant impact on throughput. In other words, managers should take into account the impact of the knowledge transferred between team workers when making grouping and assignment decisions.

Having validated the merit in recognising knowledge transfer, we next turn to assessing the capabilities of solving the RPM to develop workforce plans while considering that knowledge transfer occurs.

5.3 Benchmarking solving the RPM

We next compare the quality of the solutions produced by solving the RPM to the solutions produced by the heuristic solution methods presented in Nembhard and Bentefouet (2015). They consider a problem that consists of making three decisions: (1) selecting workers, (2) grouping workers and (3) assigning groups of workers to job groups. To solve this problem, they decompose it into three decision-making problems, and present heuristics for each (a discussion of the details of the heuristics for problems (2) and (3) is provided in Appendix 3). Thus, to benchmark RPM, we presume a given workforce (thus obviating the need to make the first set of decisions) and compare the solutions produced by RPM to those produced by their best heuristics for the second and third sets of decisions.

Specifically, let obj_N^k be the throughput produced by the k th heuristic proposed by Nembhard and Bentefouet (2015) and obj_{RPM} the throughput produced by solving the RPM. We calculate $gap = (obj_{RPM} - \max_{k=1}^{10} obj_N^k)/obj_{RPM}$. We display the distribution of these gaps in Figure 2 and note that the average of these gaps is 10.59%. We conclude that solving the RPM produces significantly higher throughput than the existing heuristics proposed. We note that for these experiments, we use the same 100 instances as in the previous comparison.

Having observed that plans prescribed by the RPM produce significantly higher throughput, we next seek to understand the scalability of a planning approach that includes solving the RPM.

5.4 Scalability of solving RPM

We next study how the time needed to solve an instance of the RPM changes as the number of workers, groups, and/or group sizes changes. We first focus on situations where workers are to be paired (i.e. groups consist of two workers). In Figure 3, we illustrate the time required to solve the RPM for different number of workers. We note that in these instances the number of tasks equals the number of workers. For each number of workers, we consider 10 instances and illustrate the average solve

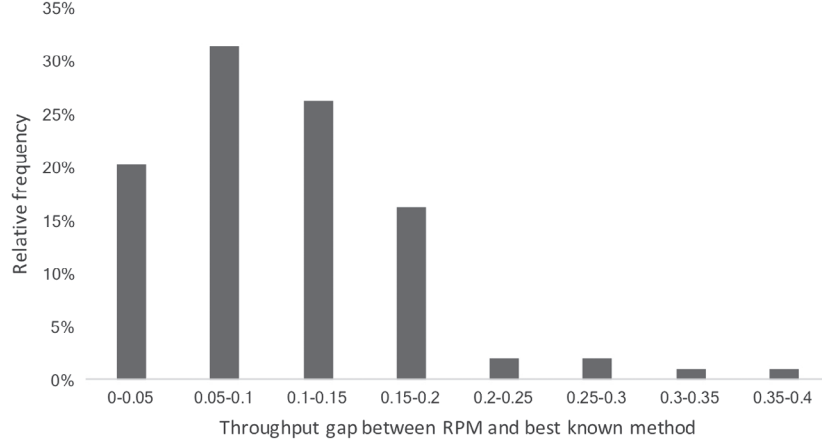


Figure 2. Distribution of throughput gaps between solving RPM and best existing heuristic.

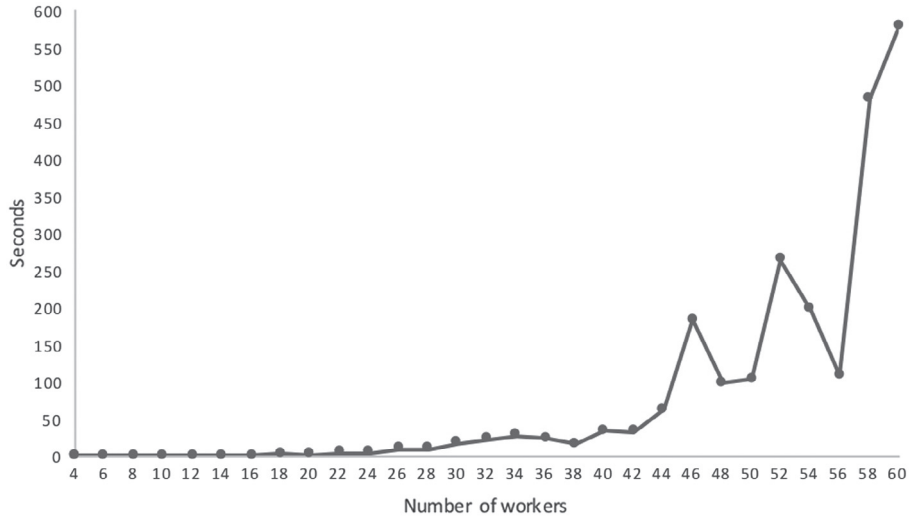


Figure 3. Time to solve RPM by number of workers when pairing workers.

time over those 10 instances. We see that the solve time grows in the number of workers, but even when putting 60 workers into pairs, instances can be solved in less than 10 min.

We next study how easily the RPM can be solved when there are larger groups (e.g. group sizes ranging from three to six). In Table 2, we report the run time required to solve instances of the RPM when workers are put into groups of size three, four, five, and six as the number of workers changes. We see that for a fixed group size, the solve time increases as the number of workers increases. We also see that for groups of size three, four, and five workers, all instances are solved to optimality within 20 min when the workforce has less than 28 people. However, there were instances wherein Gurobi exceeded the memory it was allotted before being able to solve the instance to optimality. Specifically, when considering groups of six workers, we were unable to solve instances with more than 18 workers. However, for the instances with 24 and 30 workers, Gurobi was able to produce a primal solution and dual bound that resolved to a small optimality gap.

5.5 Managerial insights

We next turn our attention to managerial insights that can be gained by analysing high-quality solutions to the RPM. Recall that the RPM determines both how workers should be grouped and which group of workers should be assigned to each group of jobs. Thus, we first focus on analysing these solutions to determine how workers should be grouped. To do so, we let σ_w^α

Table 2. Solve times when considering larger groups.

Num worker	Obj	CPU	Gap	Num worker	Obj	CPU	Gap
Group size = 3				Group size = 4			
6	199.86	0.11	0	8	287.49	0.05	0
9	322.90	0.23	0	12	491.19	0.82	0
12	448.92	0.60	0	16	658.15	6.60	0
15	579.83	2.69	0	20	855.31	17.97	0
18	714.46	10.98	0	24	1039.80	350.37	0
21	847.42	60.00	0	28	1213.63	395.02	0
24	1000.81	85.88	0	32	1,410.15	1109.35	0
27	1138.58	1179.44	0	36	1,596.69	3710.11	0.01
30	1276.70	3605.42	0	40	1,780.98	3792.01	0.01
33	1431.63	3608.71	0.004	44	1955.54	3965.85	0.02
				48	2160.22	4081.55	0.01
				52	2323.39	4326.82	0.02
Group size = 5				Group size = 6			
10	383.65	0.42	0	12	499.63	1.02	0
15	623.58	8.07	0	18	775.19	39.20	0
20	874.95	36.96	0	24	1032.50	5341.26	0.01
25	1095.87	527.50	0	30	1308.47	7087.10	0.02
35	1555.94	4277.42	0.01				
40	1573.13	5405.63	0.14				

Table 3. Average dissimilarity index.

	K	p	$1/r$	θ
Avg. dissim $_{\text{RPM}}^{\alpha}$	70.12%	74.80%	79.17%	68.83%

denote the standard deviation associated with the different values of the learning attributes α in the group of workers w . We then calculate $\sigma_{\max}^{\alpha} = \max_{w \in \mathcal{W}} \sigma_w^{\alpha}$ and $\sigma_{\min}^{\alpha} = \min_{w \in \mathcal{W}} \sigma_w^{\alpha}$. In other words, for an attribute α , $\sigma_{\max}^{\alpha}$ represents the largest standard deviation over all possible groupings of workers and $\sigma_{\min}^{\alpha}$ the smallest.

We use these values to determine the degree to which the individuals in group w are dissimilar with respect to learning attribute α . Specifically, for group w and attribute α , we calculate $\text{dissim}_w^{\alpha} = (\sigma_{\max}^{\alpha} - \sigma_w^{\alpha}) / (\sigma_{\max}^{\alpha} - \sigma_{\min}^{\alpha})$. dissim_w^{α} takes on a value between 0 and 1, with a value of 1 implying that w is a grouping of very similar individuals with respect to learning attribute α (i.e. $\sigma_w^{\alpha} = \sigma_{\min}^{\alpha}$; they are as similar as possible given the workforce) and 0 implying that w is a grouping of very dissimilar individuals. Then, given a set of groups, $\mathcal{W}_{\text{RPM}}$ prescribed by a solution to the RPM, we calculate an average dissimilarity index with respect to α , $\text{dissim}_{\text{RPM}}^{\alpha}$, as $\sum_{w \in \mathcal{W}_{\text{RPM}}} \text{dissim}_w^{\alpha} / |\mathcal{W}_{\text{RPM}}|$. We report averages of $\text{dissim}_{\text{RPM}}^{\alpha}$ over the 100 instances tested and the four learning attributes considered in Table 3 (first presented in Jin (2016)). What we conclude from this table is that it is better to group individuals who are dissimilar (as the indices are greater than 50%) than similar. We note that this is in stark contrast to how the heuristics from the literature operate, as they seek to group workers that have similar values for one of the learning attributes.

We next turn our attention to how groups of workers should be assigned to groups of jobs. To analyse this component of solutions to the RPM, consider workers $w = (w_1, w_2, \dots, w_n)$ and job-type g . Let $\alpha_{w_i}^g$ represent worker w_i 's value for attribute α on jobs of type g . We then let $\alpha_{\max}^{wg} = \max_{i=1}^n \alpha_{w_i}^g$. Then, with $\mathcal{W}_{\text{RPM}}$ representing the groups of workers prescribed by the solution to RPM and $\mathcal{G}$ the job-types in the instance we calculate $\alpha_{\max}^{wg} = \max_{w \in \mathcal{W}_{\text{RPM}}} \max_{g \in \mathcal{G}} \alpha_{\max}^{wg}$. Similarly, we calculate $\alpha_{\min}^{wg} = \min_{w \in \mathcal{W}_{\text{RPM}}} \min_{g \in \mathcal{G}} \alpha_{\max}^{wg}$. Finally, with w_g representing the group of workers assigned to job-type g in a solution to the RPM, we calculate $\alpha_{\text{RPM}}^{wg} = \max_{g \in \mathcal{G}} \alpha_{\max}^{wg}$.

A greedy assignment policy would first assign the group of workers w to the job-type g that maximised α_{wg} (or the group w_g such that $\alpha_{\max}^{wg} = \alpha_{\max}$). To measure whether solutions to the RPM exhibit the same property, we calculate a greedy index, $\text{greedy}_{\text{RPM}}^{\alpha} = (\alpha_{\text{RPM}} - \alpha_{\min}) / (\alpha_{\max} - \alpha_{\min})$. $\text{greedy}_{\text{RPM}}^{\alpha}$ will take on a value between 0 and 1, with the closer the value is

Table 4. Average greedy index.

	K	p	$1/r$
Avg. greedy $_{\text{RPM}}^{\alpha}$	85.87%	52.63%	31.18%

to 1, the more greedily the assignment stage was performed. We report averages of greedy $_{\text{RPM}}^{\alpha}$ over the 100 instances tested and three learning attributes in Table 4 (first presented in Jin (2016)).

We conclude from these results that whether the assignment of groups to job-types should be done in a greedy fashion depends on the attribute that is being considered. When focusing on steady-state productivity (K), groups should be assigned to job-types in a greedy fashion. However, when focusing on learning rate ($1/r$), a greedy approach is less effective. While not an optimal policy, we conclude that assigning groups of workers to groups of jobs in a greedy manner based on steady-state productivity is an effective strategy.

6. Conclusions and future work

In this paper, we focus on a workforce planning problem wherein the impact of knowledge transfer between individuals is explicitly recognised. As the impact of knowledge transfer is directly related to how individuals learn and as quantitative models of how individuals learn are non-linear, existing research has proposed mixed integer non-linear programming formulations of the problem. However, the non-linear nature of such a formulation renders it computationally challenging to solve, and existing research has focused on heuristic procedures for producing solutions. We overcome this challenge by characterising the structure of the optimal solution and exploiting the structure through a reformulation. This reformulation leads to our second contribution; we are able to solve, to optimality, much larger instances than existing approaches in the literature have approached with heuristic procedures. This paper presents the first approach that can solve problems up to 60 workers. Finally, analysing the structure of optimal solutions to the problem, we show that it is best to group together workers that are dissimilar with respect to learning parameters. This contribution gives managers a new insight in the field of workforce planning.

In addition to the insights provided by this paper, this paper also reveals several opportunities for future work. First, we note that the proof of Proposition 4.1 considers the case in which there are more workers than jobs. Thus, our method could also be used to select workers in addition to grouping and assigning them. In addition, this study focuses on parallel production systems. However, many production systems are serial systems. In some industries, there is a mixture of parallel and serial systems. An interesting direction of future work would be to seek to extend the proposed methodology to serial systems or more complex configurations. Lastly, while the proposed methodology solves much larger instances to optimality than has previously been seen in the literature, the proposed solution method struggles as the number of workers and/or size of worker groups becomes large. Thus, to solve larger problems, additional research is required. Notably, the enumerative nature of the reformulation suggests that a column generation-based solution approach could be effective. At the same time, one does wonder if there is a group size at which there are diminishing returns to knowledge transfer. Simply, there would be a point at which the interactions among some in the group are too few or people are too far away to lead to transfer. The answer to such a question could be the subject of future empirical research.

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Appendix 1. Problem formulation when jobs in each group can differ

In this section, we present a more general formulation of the problem than that presented in Section 3. Notably, here, we allow the jobs in each group to differ. However, we assume that learning-by-transfer is still possible due to the similar nature of jobs in a group as well as the co-location of these jobs.

To model this problem, we assume a horizon of T periods, and let the set $\mathcal{T} = \{1, 2, \dots, T\}$ index the set of periods in the horizon. We define $\mathcal{I} = \{1, 2, \dots, I\}$ to be a set of workers that perform jobs in set $\mathcal{J} = \{1, 2, \dots, J\}$. With this discrete planning horizon, we model the assignment of individuals to jobs with the binary variable x_{ij}^t , which indicates whether worker i performs job j in period t . Associated with each pair of a worker and a job is the learning parameters that determine the worker's ability to learn the job. Referring to Equation (1), we let K_{ij} , p_{ij} , and r_{ij} denote the asymptote parameter, prior experience, and learning rate of worker i on job j . The parameter r_{ij} captures the rate at which worker i learns job j by expressing how much experience the individual would need to achieve an output level of $K_{ij}/2$. We note that, with these parameters, we are explicitly modelling that the workforce is heterogeneous with respect to productivity and learning.

Jobs are partitioned into G different groups, and $\mathcal{G} = \{1, 2, \dots, G\}$ is the set of these groups. Jobs in group $g \in \mathcal{G}$ are a subset of jobs in $\mathcal{J}$ that are grouped together so that workers working them are close enough to transfer knowledge. We also assume that the ability of a worker to absorb knowledge from her/his colleagues' accumulated experience does not depend on the jobs performed. Instead, associated with each worker is a transfer parameter $\theta_i \in [0, 1]$, which represents the fraction of knowledge transferred to worker i from her/his coworkers that are working on jobs in the same group.

For each period t in the horizon, we let c_{ij}^t denote the accumulated work experience of worker i for task j up to period t . As discussed previously, the accumulated work experience c_{ij}^t is a function of direct and indirect experience. The direct experience is the number of periods in which worker i performs job j by time t , i.e. $D_{ij}^t = \sum_{t' < t} x_{ij}^{t'}$. For worker i in period t , we denote the experience of colleagues that influences i 's productivity on job j as S_{ij}^t . We call S_{ij}^t the indirect experience of worker i on job j up to time t . We calculate this as the total production output of all colleagues on jobs that are in the same group as j as follows:

$$S_{ij}^t = \sum_{i' \in \mathcal{I}: i' \neq i} \sum_{k \in \mathcal{G}: k \neq j} \sum_{t' < t} \varphi_{i'k}(c_{i'k}^{t'}), \quad i \in \mathcal{I}, j \in \mathcal{G}, g \in \mathcal{G}, t \in \mathcal{T} \quad (\text{A1})$$

Then, we calculate the accumulated experience of worker i on job j in period t as:

$$c_{ij}^t = \theta_i S_{ij}^t + D_{ij}^t, \quad i \in \mathcal{I}, j \in \mathcal{J}, g \in \mathcal{G}, t \in \mathcal{T}. \quad (\text{A2})$$

We can then generalise Equation (2) as:

$$\varphi_{ij}(c_{ij}^t) = K_{ij} \frac{c_{ij}^t + p_{ij}}{c_{ij}^t + p_{ij} + r_{ij}}, \quad i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T}. \quad (\text{A3})$$

Recall that we assume that jobs have already been partitioned into groups such that knowledge can be transferred between individuals working on jobs of the same type. As such, we model the choice of putting workers into groups (or teams), with each group of workers assigned to a single job group. We define the binary variable y_{ig}^t to indicate whether worker i is assigned to group $g \in \mathcal{G}$ in period t .

The objective is to maximise the total system production during the horizon (which we often refer to as system throughput). We let the continuous variable o_{ij}^t represent worker i 's output on job j in period t . We define the following non-linear programme for assigning workers to groups and groups of workers to job groups in a parallel system:

$$\text{Max} \quad \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \sum_{t \in \mathcal{T}} o_{ij}^t$$

s.t (A1)–(A3)

$$\sum_{i \in \mathcal{I}} x_{ij}^t \leq 1, \quad j \in \mathcal{J}, t \in \mathcal{T} \quad (\text{A4})$$

$$\sum_{g \in \mathcal{G}} y_{ig}^t = 1, \quad i \in \mathcal{I}, t \in \mathcal{T} \quad (\text{A5})$$

$$\text{(PM)} \quad \sum_{j \in \mathcal{G}} x_{ij}^t \leq y_{ig}^t, \quad i \in \mathcal{I}, g \in \mathcal{G}, t \in \mathcal{T} \quad (\text{A6})$$

$$o_{ij}^t \leq x_{ij}^t \varphi_{ij}(c_{ij}^t), \quad i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T} \quad (\text{A7})$$

$$o_{ij}^t \geq 0, \quad i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T}. \quad (\text{A8})$$

$$x_{ij}^t, y_{ig}^t \in \{0, 1\}, \quad i \in \mathcal{I}, j \in \mathcal{J}, t \in \mathcal{T}, g \in \mathcal{G} \quad (\text{A9})$$

Constraints (A4) restrict that a job can be performed by only one worker in a period. Constraints (A5) enforce that a worker is assigned to only one group in each period, while constraints (A6) specify that workers can only work on jobs within their assigned group for a

given period. Lastly, constraints (A7) indicates that the output on a job in a period is limited both by a worker's productivity and whether the worker performed that job in that period. Constraints (A8) and (A9) give the domain of decision variables. We refer to this model as 'PMM.'

We can reformulated *PMM* analogously to the reformulation for *PM*. We need only redefine O_w^g . To do so, we enumerate the possible assignment of workers to jobs in each job type. For a job-type g consisting of n jobs, we let $\bar{W}_g$ denote the set of all possible permutations of n workers, e.g. $\bar{w} = \{i_1, i_2, \dots, i_n\}$, $\bar{w} \in \bar{W}_g$. Note that, from the ordering of the individuals in $\bar{w}$, we derive the assignment of a worker in $\bar{w}$ to a job in g . In other words, with $\bar{w} = \{i_1, i_2, \dots, i_n\}$ and $g = \{j_1, j_2, \dots, j_n\}$, we assign i_1 to perform j_1 , i_2 to perform j_2 , etc. Then, given $\bar{w}$ and g , we calculate the throughput of this group of workers $\bar{w}$ when assigned to group g , which we denote as $O_{\bar{w}}^g$, as follows:

$$\begin{aligned} \bar{O}_{\bar{w}}^g &= \sum_{l=1}^n o_{i_l^{\bar{w}} j_l} \\ &= \sum_{l=1}^n \sum_{t \in \mathcal{T}} K_{i_l^{\bar{w}} j_l} \frac{c_{i_l^{\bar{w}} j_l}^t + p_{i_l^{\bar{w}} j_l}}{c_{i_l^{\bar{w}} j_l}^t + p_{i_l^{\bar{w}} j_l} + r_{i_l^{\bar{w}} j_l}} \\ &= \sum_{l=1}^n \sum_{t \in \mathcal{T}} K_{i_l^{\bar{w}} j_l} \frac{t + p_{i_l^{\bar{w}} j_l} + \sum_{q=1:q \neq l}^n \sum_{r=1:r \neq l}^n \sum_{t' < t} \varphi_{i_q^{\bar{w}} j_r} (c_{i_q^{\bar{w}} j_r}^{t'})}{t + p_{i_l^{\bar{w}} j_l} + \sum_{q=1:q \neq l}^n \sum_{r=1:r \neq l}^n \sum_{t' < t} \varphi_{i_q^{\bar{w}} j_r} (c_{i_q^{\bar{w}} j_r}^{t'}) + r_{i_l^{\bar{w}} j_l}}. \end{aligned} \quad (\text{A10})$$

Appendix 2. Proof of Proposition 4.1

In this section, we present the proof of Proposition 4.1.

Proof. By contradiction.

We start our proof with the case that the number of workers is equal to the number of jobs, $I = J$. In this case, we first consider a system of two workers and two tasks in the time horizon $\mathcal{T}$ with T periods. We assume a specialised scheduling s is the best scheduling from all specialised scheduling that yields the maximum system throughput over the horizon. We prove that this specialised scheduling is the optimal solution for the system.

In a parallel system with two workers and two jobs, we denote the best specialised scheduling as s^* , which assigns worker 1 to job 1 and worker 2 to job 2. The system throughput of the specialised scheduling s^* is:

$$O_{s^*} = \sum_{k \in \mathcal{T}} \varphi_{11} (c_{11}^k) + \sum_{k \in \mathcal{T}} \varphi_{22} (c_{22}^k) \quad (\text{B11})$$

Since s^* is the best specialised scheduling, the other specialised scheduling of assigning worker 1 to job 2 and worker 2 to job 1 has a smaller or equal system throughput than s^* , which says:

$$\sum_{k \in \mathcal{T}} \varphi_{12} (c_{12}^k) + \sum_{k \in \mathcal{T}} \varphi_{21} (c_{21}^k) \leq \sum_{k \in \mathcal{T}} \varphi_{11} (c_{11}^k) + \sum_{k \in \mathcal{T}} \varphi_{22} (c_{22}^k). \quad (\text{B12})$$

Assume there exists a non-specialised scheduling ns that yields a higher system throughput than s^* , which implies:

$$O_{s^*} < O_{ns}. \quad (\text{B13})$$

In scheduling ns , there exists at least one switch time point. We say the switch point is t , $t > 1$. Then, the number of time periods that a worker is assigned to a job is as follows:

- Worker 1 is assigned Job 1 t times and Job 2 $T - t$ times
- Worker 2 is assigned Job 1 $T - t$ times and Job 2 t times

Therefore, the total system throughput for the non-specialised scheduling ns is:

$$\begin{aligned} O_{ns} &= \sum_{k \leq t} \varphi_{11} (c_{11}^k) + \sum_{k > t} \varphi_{12} (c_{12}^k) + \sum_{k > t} \varphi_{21} (c_{21}^k) + \sum_{k \leq t} \varphi_{22} (c_{22}^k) \\ &= \sum_{k \leq t} \varphi_{11} (k - 1 + \theta_1 \varphi_{22} (c_{22}^k)) + \sum_{k > t} \varphi_{12} (k - t - 1 + \theta_1 \varphi_{21} (c_{21}^k)) \\ &\quad + \sum_{k \leq t} \varphi_{22} (k - 1 + \theta_2 \varphi_{11} (c_{11}^k)) + \sum_{k > t} \varphi_{21} (k - t - 1 + \theta_2 \varphi_{12} (c_{12}^k)) \end{aligned} \quad (\text{B14})$$

To analyse the knowledge transferred from coworkers in the same team, we assume that the knowledge/experience gained from coworkers in a given period can never exceed the experience gained if the worker was assigned to the task j itself during the same period. We add the constraints that:

$$S_{ij}^t - S_{ij}^{t-1} = \sum_{p \in \mathcal{T}} \sum_{q \in g}^{p \neq i} \varphi_{pq} (c_{pq}^t) < 1. \quad (\text{B15})$$

Constraints (B15) enforce that knowledge transferred from team members' experience on other jobs could not result in a learning for job j superior to the learning the worker would have reached if the worker is assigned to j during the same time period. Then, Equation (B11) can be expanded as the following way:

$$\begin{aligned}
O_{s^*} &= \sum_{k \leq t} \varphi_{11}(c_{11}^k) + \sum_{k > t} \varphi_{11}(c_{11}^k) + \sum_{k > t} \varphi_{22}(c_{22}^k) + \sum_{k \leq t} \varphi_{22}(c_{22}^k) \\
&= \sum_{k \leq t} \varphi_{11}(k-1 + \theta_1 \varphi_{22}(c_{22}^k)) + \sum_{k > t} \varphi_{11}(k-1 + \theta_1 \varphi_{22}(c_{22}^k)) \\
&\quad + \sum_{k > t} \varphi_{22}(k-1 + \varphi_{11}(c_{11}^k)) + \sum_{k \leq t} \varphi_{22}(k-1 + \varphi_{11}(c_{11}^k)) \\
&\text{By (B12):} \\
&> \sum_{k \leq t} \varphi_{11}(k-1 + \theta_1 \varphi_{22}(c_{22}^k)) + \sum_{k > t} \varphi_{12}(k-1 + \theta_1 \varphi_{22}(c_{22}^k)) \\
&\quad + \sum_{k > t} \varphi_{22}(k-1 + \varphi_{11}(c_{11}^k)) + \sum_{k \leq t} \varphi_{21}(k-1 + \varphi_{11}(c_{11}^k)) \\
&\text{By (B15) and } t > 1 : \\
&> \sum_{k \leq t} \varphi_{11}(k-1 + \theta_1 \varphi_{22}(c_{22}^k)) + \sum_{k > t} \varphi_{12}(k-t-1 + \theta_1 \varphi_{21}(c_{21}^k)) \\
&\quad + \sum_{k \leq t} \varphi_{22}(k-1 + \theta_2 \varphi_{11}(c_{11}^k)) + \sum_{k > t} \varphi_{21}(k-t-1 + \theta_2 \varphi_{12}(c_{12}^k)) \\
&= O_{ns}.
\end{aligned} \tag{B16}$$

This shows that $O_{s^*} > O_{ns}$, which contradicts to the assumption in (B13). Same contradictions can be achieved when there are two or more switch points. Therefore, we prove that the best specialised scheduling s^* yields the highest system throughput and therefore is the optimal scheduling.

The proof for the case in which $I \neq J$ proceeds in a similar fashion. We first address the case of $I < J$, where the number of workers is fewer than the number of jobs. We let $N = J - I$ denote the number of surplus jobs compared to the scenario of equal size of workers and jobs in Proposition 1. Thus, $\bar{I} = I + N$ is equal to the number of jobs J , where workers $\{1, \dots, I\}$ represent actual workers, and workers $\{I+1, \dots, J\}$ represent an additional requirement, termed artificial workers.

For $i \leq I$, the initial distribution of worker's skills and performance is maintained. For $I < i \leq J$, constraints (A7) become:

$$o_{ij}^t \leq \varepsilon \Phi_{ij}^t(c_{ij}^t), \forall i = I+1, \dots, J, j \in \mathcal{J}, t \in T \tag{B17}$$

Next, we consider the case of $I > J$. The number of workers is greater than the number of jobs. We let $M = I - J$ denote the number of surplus workers. Thus, using a similar approach as above, M artificial stations are introduced, such that the number of jobs $\bar{J} = J + M$ is equal to the number of workers I , where the 'real jobs' are $j \in \{1, \dots, J\}$ and artificial jobs are $j \in \{J+1, \dots, I\}$.

For $j \leq J$, the initial distribution of worker's skills and performance on task j is maintained. For $J < j \leq I$, constraints (A7) become:

$$o_{ij}^t \leq \varepsilon \Phi_{ij}^t(c_{ij}^t), \forall j \in \{J+1, \dots, I\}, i \in \mathcal{I}, t \in T \tag{B18}$$

ε is chosen to be a small number that the contribution of the artificial workers to any task will be negligible. A good choice would be such that $\varepsilon \times \max\{\sum_{i \in \mathcal{I}} \Phi_{ij}^t(c_{ij}^t), \forall i \in \mathcal{I}, j \in \mathcal{J}\}$ is negligible compared to 1. Therefore, when $I < J$, the workers in $\mathcal{I}$ are allocated to the most productive of the jobs in $\mathcal{J}$. The remaining jobs are in vacancy. When $I > J$, scheduling workers to switch between jobs at any time is suboptimal. That is, workers should be assigned to a single task during the entire period to maintain optimal production. The result also indicates that the M surplus workers will not be used during the time horizon for the current schedule. $\square$

Appendix 3. Grouping and assignment policies presented in Nemhard and Bentefouet (2015)

The top 10 policies for parallel systems presented in Nemhard and Bentefouet (2015) considering only the grouping and assignment stages are listed in Table C1.

Nemhard and Bentefouet (2015) do not explain how each policy is calculated. We contacted the authors and received the explicit form for the policies. For example, Nemhard and Bentefouet (2015) assume nine workers are selected. Then, Min variance K uses the following procedure. Nemhard and Bentefouet (2015) considered systems with three types of machine groups, each with three identical jobs. Workers have common learning and transfer parameters for each machine type. Assume the three machine types are a, b , and c . Each worker i has three values for the parameters. For example, K has values of K_{ia}, K_{ib}, K_{ic} . The average K value of worker i is $K_i = (K_{ia} + K_{ib} + K_{ic})/3, i = 1, \dots, 9$. The grouping policy of Min variance K is a way of grouping the nine workers to three groups such that the total variance of the three groups is the smallest. The variance of a group with workers i_1, i_2, i_3 is: $((k_{i_1} - \mu)^2 + (k_{i_2} - \mu)^2 + (k_{i_3} - \mu)^2)/3$, where μ is the average of the three workers k values. The assignment policy assigns each team to a job group. A policy of Maximax K assigns the job group to the team whose maximal K value on this job is the highest over the three teams. The other grouping and assignment policies follow the same fashion.

Table C1. Top 10 policies in Nembhard and Bentefouet (2015).

Rank	Parallel system	
	Grouping	Assignment
1	Min variance K	Maximax K
2	Min variance p	Maximax θ
3	Min variance K	Maximax θ
4	Min variance $1/r$	Maximax K
5	Min variance $1/r$	Maximax θ
6	Min variance p	Maximax K
7	Min variance θ	Maximax θ
8	Min variance θ	Maximax K
9	Min variance O	Maximax p
10	Min variance O	Maximax p